NIST base points and the B-163 seed: generator searches

2026-10-17 (records), 2026-10-22 (this report). Directory `research/nist_basepoint_provenance_20261017` in `aburan28/crypto`.

Status. Measured negative result. No tested generator, library or derivation reproduces the base points of P-224, P-256, P-384, P-521, the five B-curves or the five K-curves, or the B-163 seed. This concerns the provenance of published constants. It has no bearing on ECDLP hardness: any point of order n is an equally good generator.

Question. The six May-1997 base points fit FreeLIP `zrandom`. The other four prime-curve base points, every binary-curve base point and the B-163 seed remained unexplained (`nist_seed_provenance_20261016`). The standards never say how G was chosen. FIPS 186-4 D.1.1.5 says "any point of order n can serve". X9.62-1998 A.3.1 draws R at random and sets $G = hR$.

Binary-curve correction. For $h = 2$ or 4, the random draw is $x(R)$, not G_x. Every binary target therefore includes all R with $hR = G$, in both the polynomial and the normal basis. The basis conversion was checked first: the normal-basis b equals the SHA-1 seed output for all five B-curves, and each halving doubles back to G. There are 137 binary targets in all.

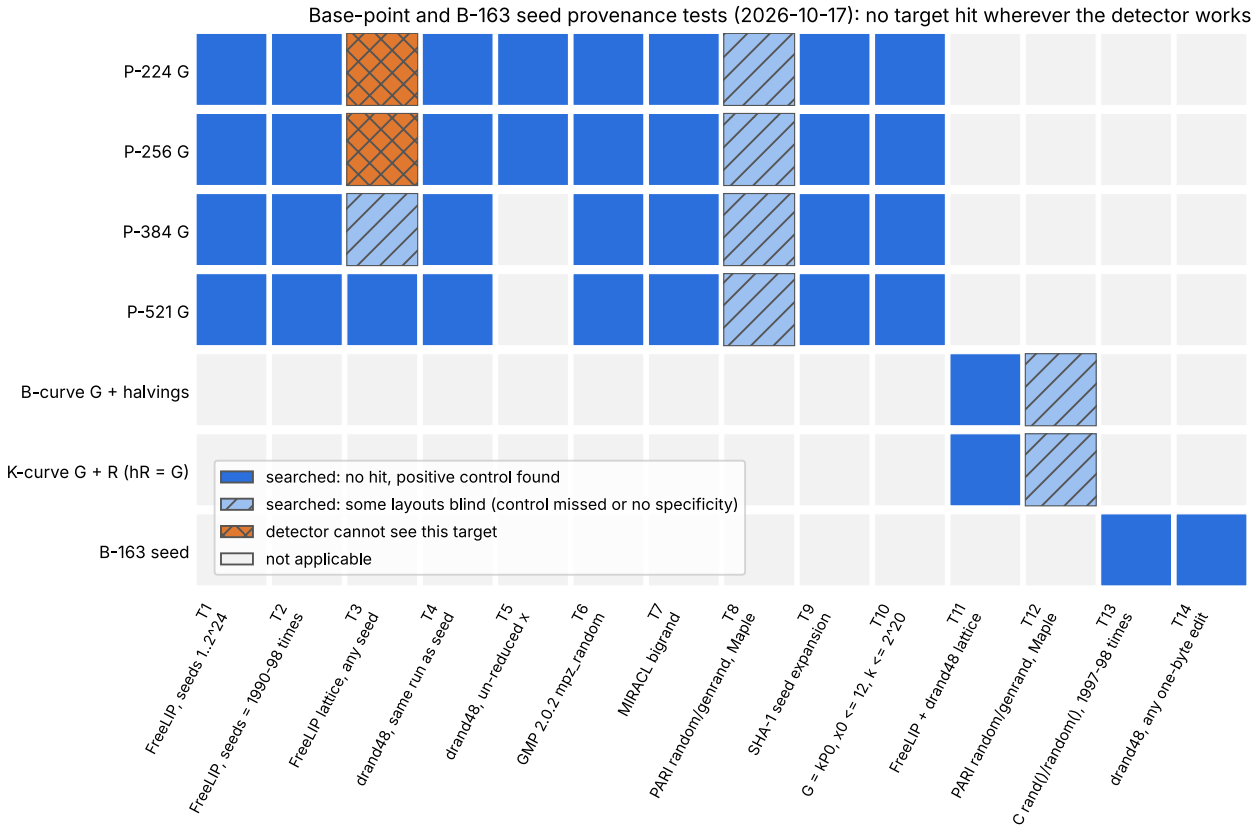


Figure 1: Search coverage. Blue: searched, no target hit, positive control found. Light hatched: some layouts blind. Orange cross: the detector cannot see the target. Drawn from `out/coverage.json`, transcribed from the `out/*.out` records.

Searches (all controls found; 0 target hits)

id	search and scope	targets
T1	FreeLIP `zrandomb`, seeds $1…2^{24}$, NBITS 26/30/62, offsets 0–255	P-224…P-521 G_x
T2	FreeLIP, seed = Unix time 1990–1998, offsets 0–63	same
T3	FreeLIP lattice, any seed (blind on P-224/P-256 and on P-384 at NBITS 62)	G_x, G_y
T4	drand48 continuing the published seed run	P-224…P-521
T5	drand48 chunks, un-reduced $x = X \bmod p$	P-224, P-256
T6	GMP 2.0.2 `mpz_random` on glibc/4.3BSD `random()`, seeds $0…2^{24}$ and 1997–98 times	P-curves
T7	MIRACL `bigrand`, two parameter sets, default state and `irand` seeds	P-curves
T8	PARI `random()`/genrand (5 packings); Maple V `rand` (control missed)	P-curves

T9	SHA-1 expansion of the curve seed, counters $0 \ldots 2^{16}$	P-curves
T10	$G = kP_0$, $x_0 \le 12$, $k \le 2^{20}$	P-curves
T11	FreeLIP and drand48-family lattice scan	137 binary
T12	PARI / Maple patterns	137 binary
T13	glibc/4.3BSD `random()`, SysV/BSD/MSVC `rand()`, seeds 1997–98 t, $t/2$	B-163 seed
T14	drand48 family, every short window: covers any one-byte edit	B-163 seed

Side test. A preregistered CM-discriminant screen of the skip filter found no split. Of the published candidates, 8 of 10 have $f = 1$ (the other two have $f = 5$ and $f = 3$); of the skipped, 10 of 11 do. It is superseded by H7 in `nist_seed_skip_filter_20261017`.

Caveats. Wall times were not recorded. The invocations in `run.sh` are reconstructed from each record's "done:" line; T9 and T12 were re-run and match byte for byte. Magma, SIMATH, KANT, ZEN/BigNum and libI were not tested. There has been no independent review.

What this leaves. None of these base points or the B-163 seed is explained. Either the generator is one not covered here, or the choice is not a function of a recoverable seed. Neither can be tested from public data.